

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}.\frac{1}{315}.$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

Simplify Multipoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{vmatrix} +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \end{vmatrix} \quad (4a)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot$$

$$\begin{aligned}&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta}\\&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma}\\&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma}\\&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta}\\&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma}\\&+105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&+105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha}\\&-15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon}\end{aligned}$$

(5a)

Reduce Indices:

$$\begin{aligned} T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot & \\ & +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ & +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ & +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ & +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ & +105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & +105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ & -15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{aligned} \quad (6a)$$

Add up:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} -525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +315 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (7)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = \begin{bmatrix} -\frac{5}{3} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -\frac{10}{3} \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ -5 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +1 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{\alpha\alpha\beta\beta} \\ +4 \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \\ -\frac{4}{7} \cdot R_z^{-7} \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\beta\beta} \end{bmatrix} \quad (8)$$